

eduAssets — Raspberry Pi 5 in Kiosk Mode

Ubuntu Desktop • PostgreSQL (LAN accessible) • Express/Prisma Backend • Nginx • Chromium Kiosk

1. Base System

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y git curl build-essential nginx postgresql ufw
sudo reboot
```

2. Node.js

The `frontend/package-lock.json` locks `vite@8.2.0`, whose engines require node `^20.19.0 || >=22.12.0`. Use NodeSource, not the default apt package:

```
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt install -y nodejs
node --version      # v22.x
```

3. Chromium (Actual Installation)

On modern Ubuntu Desktop (22.04+), `chromium` / `chromium-browser` from apt is a transitional package that pulls the snap. The direct and reliable way is to install the snap explicitly:

```
sudo apt install -y snapd
sudo snap install chromium
```

Verify:

```
which chromium
chromium --version
```

The binary is located at `/snap/bin/chromium`, which is already in the PATH by default in Ubuntu graphical sessions. If `which chromium` finds nothing, open a new terminal/session (the snap PATH is configured via `/etc/profile.d/apps-bin-path.sh`, which is only read on a new login).

4. Clone and Install

(monorepo: `backend/` + `frontend/`, no `package.json` in the root)

```
mkdir -p ~/Projects && cd ~/Projects
git clone YOUR_REPOSITORY_URL eduassets
cd eduassets/backend && npm install
cd ../frontend && npm install
```

5. PostgreSQL — Install and Expose to LAN (Restricted)

5.1 Create User and Database

```
sudo -u postgres psql
```

```
CREATE USER eduassets WITH PASSWORD 'STRONG_PASSWORD_HERE';
CREATE DATABASE eduassets OWNER eduassets;
\q
```

5.2 listen_addresses — Restricted Configuration

`listen_addresses = '*'` makes Postgres listen on all interfaces. To expose only what is necessary, list `localhost` and the LAN static IP:

```
sudo -u postgres psql -c "SHOW config_file;"
```

Edit the returned file:

```
listen_addresses = 'localhost,192.168.1.50'
```

5.3 pg_hba.conf — Restrict to Local Subnet

In the same directory as `postgresql.conf`:

```
host      eduassets      eduassets      192.168.1.0/24      scram-sha-256
```

```
sudo systemctl restart postgresql
sudo systemctl enable postgresql
```

5.4 UFW — Allow 5432 Only for Subnet

```
sudo ufw allow from 192.168.1.0/24 to any port 5432 proto tcp
```

This applies regardless of Nginx (step 10). Postgres remains accessible directly on port 5432 for the LAN.

5.5 Static IP for Raspberry Pi

```
hostname -I
```

Reserve this IP in the router's DHCP (e.g., `192.168.1.50`).

5.6 Special Characters in the Password URL

Generate the encoded version instead of trying to escape it manually:

```
python3 -c "import urllib.parse,sys; print(urllib.parse.quote(sys.argv[1], safe=''))" 'S3nh@Forte#2026'
```

Example: `S3nh@Forte#2026` → `S3nh%40Forte%232026`.

```
postgresql://eduassets:S3nh%40Forte%232026@localhost:5432/eduassets
```

6. Backend Environment Variables

`backend/.env`:

```
cd ~/Projects/eduassets/backend  
nano .env
```

```
DATABASE_URL="postgresql://eduassets:ENCODED_PASSWORD@localhost:5432/eduassets"
DIRECT_URL="postgresql://eduassets:ENCODED_PASSWORD@localhost:5432/eduassets"

# Required (minimum 32 characters)
JWT_SECRET="PASTE_OPENSSL_RAND_RESULT_HERE"

PORT=3000

# Comma-separated list
CORS_ORIGIN="http://192.168.1.50"

SEED_ADMIN_PASSWORD="ADMIN_PASSWORD"
```

To generate the JWT_SECRET:

```
openssl rand -base64 48
```

7. Prisma — Migrations and Seed

```
cd ~/Projects/eduassets/backend
npx prisma generate
npx prisma migrate deploy
npx prisma db seed
```

8. Test Backend Manually

```
npx tsx src/server.ts
```

In another terminal:

```
curl http://localhost:3000/categorias
```

9. Frontend — Relative Origin and Deploy

9.1 Adjust API URL

```
frontend/src/core/api/apiConfig.ts :
```

```
export const API_BASE_URL: string = '';
```

9.2 Publication Directory

```
sudo mkdir -p /var/www/eduassets  
sudo chown -R $USER:www-data /var/www/eduassets  
sudo chmod -R 755 /var/www/eduassets
```

9.3 Build and Deploy

```
cd ~/Projects/eduassets/frontend  
npm run build  
sudo rm -rf /var/www/eduassets/*  
sudo cp -r dist/* /var/www/eduassets/  
sudo chown -R www-data:www-data /var/www/eduassets
```

10. Nginx — Static Files + Proxy

```
sudo nano /etc/nginx/sites-available/eduassets
```

```
server {  
    listen 80;  
    server_name 192.168.1.50;  
  
    root /var/www/eduassets;  
    index index.html;  
  
    location / {  
        try_files $uri $uri/ /index.html;  
    }  
  
    location ~ ^/(auth|categorias|equipamentos|responsaveis|usuarios|emprestimos|  
ocorrencias)(/|$) {  
        proxy_pass http://127.0.0.1:3000;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
    }  
}
```

```
sudo rm -f /etc/nginx/sites-enabled/default
sudo ln -s /etc/nginx/sites-available/eduassets /etc/nginx/sites-enabled/
sudo nginx -t && sudo systemctl reload nginx
sudo systemctl enable nginx
sudo ufw allow 80/tcp
```

11. systemd — Backend

```
sudo nano /etc/systemd/system/eduassets-backend.service
```

```
[Unit]
Description=eduAssets Backend (Express + Prisma)
After=network-online.target postgresql.service
Wants=network-online.target
Requires=postgresql.service

[Service]
Type=simple
User=YOUR_USER
WorkingDirectory=/home/YOUR_USER/Projects/eduassets/backend
EnvironmentFile=/home/YOUR_USER/Projects/eduassets/backend/.env
ExecStart=/usr/bin/npx tsx src/server.ts
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable --now eduassets-backend
sudo systemctl status eduassets-backend
```

12. Real PostgreSQL Test from Another LAN PC

```
psql -h 192.168.1.50 -U eduassets -d eduassets
```

Read:

```
SELECT id, nome FROM "Categoria" ORDER BY id;
```

Write:

```
INSERT INTO "Categoria" (nome) VALUES ('LAN Test');
```

Cleanup:

```
DELETE FROM "Categoria" WHERE nome = 'LAN Test';
```

13. Chromium Kiosk

```
mkdir -p ~/.local/bin  
nano ~/.local/bin/eduassets-kiosk.sh
```

```
#!/bin/bash  
sleep 10  
chromium --kiosk --start-fullscreen --no-first-run --disable-session-  
crashed-bubble --disable-infobars --check-for-update-interval=31536000  
"http://192.168.1.50/" &
```

```
chmod +x ~/.local/bin/eduassets-kiosk.sh
```

14. Auto-login, Auto-start, and Suspension

Auto-start via `gnome-session-properties` : * Command: `/home/YOUR_USER/.local/bin/eduassets-kiosk.sh`

Suspension:

```
echo $XDG_SESSION_TYPE
```

- **x11:** `xset s off && xset -dpms && xset s noblank`
- **wayland:** disable in Settings → Power

15. Frontend External Dependencies

Attention: Google Fonts, Material Symbols, flatpickr, SheetJS, and jsPDF depend on an active internet connection.

16. Troubleshooting

Symptom	Verification
Backend crashes on startup	<code>journalctl -u eduassets-backend -e</code> (check JWT_SECRET and DB password encoding)
Frontend does not fetch data	Confirm <code>API_BASE_URL = ''</code> and if the build/copy was redone
Nginx serves a 404 page	Check if <code>/var/www/eduassets</code> has the content and proper permissions
502/504 on Nginx	Backend active? Test <code>curl http://127.0.0.1:3000/categorias</code> locally
Login fails	Confirm if the seed script ran with <code>SEED_ADMIN_PASSWORD</code> defined
Another PC cannot access HTTP	<code>ufw allow 80/tcp</code> , IP reserved, Nginx has no errors
Another PC cannot connect to Postgres	<code>listen_addresses</code> , <code>pg_hba.conf</code> , UFW <code>5432</code> allowed
Another PC doesn't see tables	Use double quotes: <code>"Categoria"</code> , <code>"Equipamento"</code>
<code>which chromium</code> fails	<code>sudo snap install chromium</code> , open a new terminal

17. Checklist

- [x] Node 22.x via NodeSource
- [x] Chromium installed via snap
- [x] npm install in backend/ and frontend/
- [x] PostgreSQL configured for LAN + UFW
- [x] Raspberry Pi IP reserved in DHCP

- [x] Postgres password properly handled (percent-encoded)
- [x] Backend `.env` completely filled
- [x] Prisma generate, migrate, and seed executed
- [x] `apiConfig.ts` with `API_BASE_URL = ''`
- [x] Frontend build copied to `/var/www/eduassets`
- [x] Nginx configured with static routes and proxy
- [x] `eduassets-backend` service active and enabled
- [x] Postgres test via another PC successfully completed
- [x] Frontend browser access via another PC successful
- [x] Kiosk script created and auto-start configured
- [x] Screen suspension disabled